3.2.5 Coterminal angles

One complete rotation of a ray in the anticlockwise direction results in an angle measuring of $360°$. By continuing the anticlockwise rotation, angles larger than $360°$ can be produced. If we rotate in clockwise direction, negative angles are produced. Angles $57°$, $417°$ and $-303°$ have the same initial side and terminal side but with different amount of rotations, such angles are called **coterminal angles**. Thus, angles in standard position that have the same terminal sides are coterminal angles . Hence, if α and β are coterminal angles, then $\beta = \alpha + k(360°)$, k is an integer. The measurements of coterminal angles differ by an integral multiple of $360°$.

For example, $417°$ and $-303°$ are coterminal because $417° - (-303°) = 720° = 2\,(360°)$.

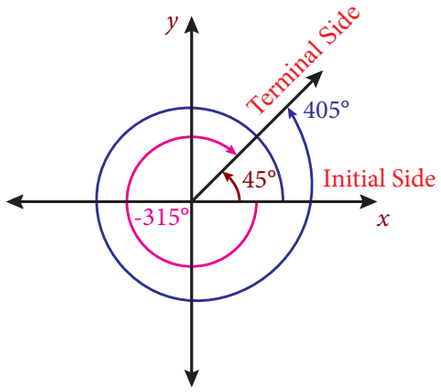


Figure 3.3

(i) Observe that $45°, -315°$ and $405°$ lie in the first quadrant.
(ii) The following pairs of angles are coterminal angles $(30°, 390°); (280°, 1000°)$ and $(-85°, 275°)$.

3.2.6 Basic Trigonometric ratios using a right triangle

We know that six ratios can be formed using the three lengths a, b, c of sides of a right triangle ABC. Interestingly, these ratios lead to the definitions of six basic trigonometric functions.

First, let us recall the trigonometric ratios which are defined with reference to a right triangle.

$$\sin\theta = \frac{\text{opposite side}}{\text{hypotenuse}} \quad ; \quad \cos\theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

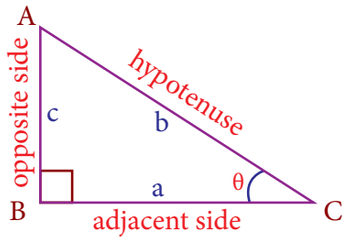


Figure 3.4

With the help of $\sin\theta$ and $\cos\theta$, the remaining trigonometric ratios $\tan\theta$, $\cot\theta$, $\text{cosec}\,\theta$ and $\sec\theta$ are determined by using the relations

$$\tan\theta = \frac{\sin\theta}{\cos\theta}\,, \quad \text{cosec}\,\theta = \frac{1}{\sin\theta}\,, \quad \sec\theta = \frac{1}{\cos\theta}\,, \quad \cot\theta = \frac{\cos\theta}{\sin\theta}\,.$$

3.2.7 Exact values of trigonometric functions of widely used angles

Let us list out the values of trigonometric functions at known angles.

θ	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

(i) The values given above are all exact.
(ii) We observe that $\sin 30°$ and $\cos 60°$ are equal. Also $\sin 60°$ and $\cos 30°$ are equal.